

SAMARTH VINAYAKA

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Education

University of Florida

Master of Science in Computer Science: CGPA - 3.9/4

August 2025 - May 2027

Gainesville, FL

Jaypee Institute of Information Technology

Bachelor of Technology in Computer Science

August 2021 - June 2025

Noida, India

Technical Skills

Languages: Python, C/C++, HTML/CSS, JavaScript, SQL

Frameworks: Pandas, NumPy, PyTorch, TensorFlow, HuggingFace, Transformers, Langchain, LangGraph, Scikit-Learn, FastAPI, Django, React, Azure, AWS, YOLO (v8/v9), Vision Language Models (VLMs), OCR, Image Processing

Skills: Data structures and Algorithm, Machine Learning, Natural Language Processing, Generative AI

Experience

Profecia Links

June 2024 – July 2024

Software Engineer Intern

Pune, India

- Engineered a natural language → SQL system over 20+ enterprise schemas using RAG (VannaAI, PandasAI), enabling non-technical users to query databases with 95%+ execution-valid accuracy (evaluated on curated query sets).
- Integrated Mistral-7B for intent classification and schema grounding, reducing query ambiguity and improving latency by 40% (5s → 3s) through optimized context construction.
- Deployed a scalable FastAPI service with modular query pipelines, request validation (Pydantic), and Dockerized workflows, enabling reliable API integration across internal systems.

India Today Group

March 2024 – May 2024

Data Science Intern

New Delhi, India

- Designed and implemented ElectionGPT, an AI chatbot utilizing GPT-4 for granular data queries on Indian elections.
- Enhanced model performance by 30% through chain of thought and few-shot prompting techniques, and built a Retrieval-Augmented Generation (RAG) pipeline for improved data retrieval and response generation.

Kalinga Institute of Industrial Technology

April 2023 – May 2024

Research Intern

Remote

- Led ABSA work for low-resource Indian languages: built the data pipeline (scraping/normalization/code-mix handling/segmentation); Proposed augmentation/preprocessing tweaks that lifted accuracy around 10% over baseline.
- Developed an augmentation pipeline (back-translation + T5 paraphrasing) with a similarity filter; curated and re-annotated around 1.3k synthetic samples to widen coverage.
- Trained and fine-tuned Hugging Face models (XLM-R, IndicBERT) and baselines (ABSA-BERT, ATAE-LSTM), achieving up to F1 98.97 (ATE) and 74.97 (APC) with XLM-R, and 69.82 (APC) with IndicBERT.

Research Publications

StressMLIoT: IoT Sensor Features Reduction and Machine Learning driven Stress Identification.

Second International Conference on Informatics (ICI)

- Architected an end-to-end ML pipeline on the SWELL-KW multi-sensor dataset using Random Forest, KNN, and SVM to classify workplace stress (up to 99% accuracy), and applied PCA for dimensionality reduction, reducing execution time by 50–75% while maintaining accuracy within 5–7%.

Projects

Self-Correcting RAG Pipeline for Reliable AI Systems

- Developed a self-correcting RAG pipeline using LangGraph with multi-stage retrieval, document grading, and query rewriting, improving answer relevance by 25–35% on ambiguous queries.
- Designed a confidence-based evaluation layer (0–1 scoring) to detect low-quality retrieval and trigger fallback strategies (query reformulation + web search), reducing hallucinated responses by 30%.
- Integrated vector search with cross-encoder re-ranking and engineered a graph-based orchestration pipeline, improving retrieval precision by 20% while maintaining latency under 2.5s across 100+ queries/day.

AutoEvolveML: Self-Evolving Multi-Agent MLOps System

- Implemented a self-evolving ML system with multi-agent control that continuously trains, evaluates, and deploys models using streaming data and feedback loops, improving model performance by 20%+ (ROC-AUC/F1) across different datasets.
- Architected an agent-driven MLOps pipeline with automated drift detection and retraining triggers, enabling continuous model adaptation and reducing performance degradation by 35% in non-stationary environments.